

Linux Mint: 24-bit / 192 kHz Audio & Dolby Atmos

System findings

Audio codec	Realtek ALC897
Linux audio server	PipeWire 1.0.5 + WirePlumber
ALSA support	32, 44.1, 48, 88.2, 96 and 192 kHz
Current PipeWire rate	48 kHz
Current allowed rates	[48000]

1. Verify PipeWire

```
pactl info | grep "Server Name"

pactl list sinks | grep -E "Name:|Sample Specification:"

aplay -l
```

The expected setup uses PipeWire and the ALC897 Analog output. The current configuration was operating at 48 kHz even though ALSA exposes 192 kHz.

2. Configure PipeWire to allow high-resolution rates

Create the configuration directory:

```
mkdir -p ~/.config/pipewire/pipewire.conf.d
```

Create a configuration file:

```
nano ~/.config/pipewire/pipewire.conf.d/10-rates.conf
```

Add:

```
context.properties = {
    default.clock.rate = 48000
    default.clock.allowed-rates = [ 44100 48000 88200 96000 192000 ]
}
```

Save in nano with Ctrl+O, Enter, then Ctrl+X. This allows PipeWire to use common sample rates instead of forcing everything to 192 kHz.

3. Restart PipeWire

```
systemctl --user restart pipewire pipewire-pulse wireplumber

pw-metadata -n settings
```

The metadata should show the allowed rates including 44100, 48000, 88200, 96000 and 192000.

4. Verify 192 kHz playback

Play a genuine 24-bit/192 kHz FLAC or WAV file in a player such as Rhythmbox or Strawberry.

```
pactl list sinks | grep "Sample Specification:"

pw-top
```

When the output path is operating at 192 kHz, the sink should report a 192000 Hz rate. Note that allowing rates does not guarantee every application or playback path will switch rates automatically.

5. If PipeWire still reports only [48000]

```
grep -R "allowed-rates\|clock.rate" ~/.config/pipewire /etc/pipewire 2>/dev/null
```

Do not edit system configuration blindly. This command identifies another configuration that may be restricting PipeWire to 48 kHz.

6. Dolby Atmos on Linux Mint

Dolby Atmos for Headphones / Windows Dolby Access: The proprietary Windows Dolby processing/driver is not directly available on Linux Mint. Installing a normal Linux audio package does not reproduce the Windows Dolby Access implementation.

Actual Dolby Atmos through HDMI: HDMI can carry Dolby formats, but the receiving TV, soundbar or AV receiver must support the relevant Atmos format and the Linux playback chain must deliver a compatible bitstream. A normal motherboard analog output cannot carry an Atmos object-based bitstream to speakers.

Virtual surround for headphones: For headphones, Linux alternatives include PipeWire/EasyEffects and other spatial/virtual-surround DSP solutions. These are not the same as Dolby Atmos for Headphones.

Important distinction

24-bit/192 kHz	Audio resolution/sample rate
Dolby Atmos	Object-based spatial audio format/processing
Virtual surround	DSP processing that simulates spatial positioning

Current hardware path: Realtek ALC897 Analog → PipeWire → ALSA. Your ALSA output confirms 192 kHz is exposed by the driver.